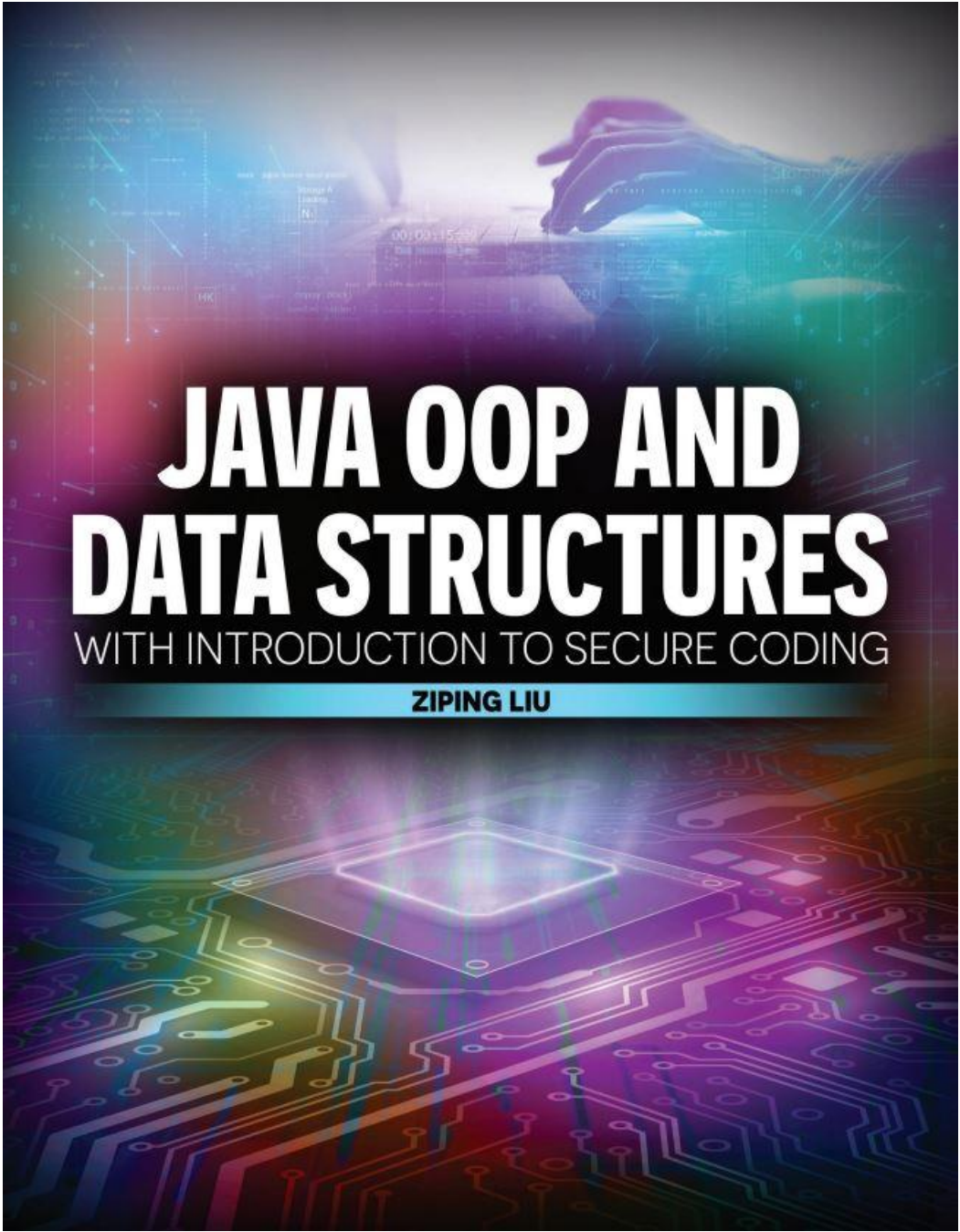
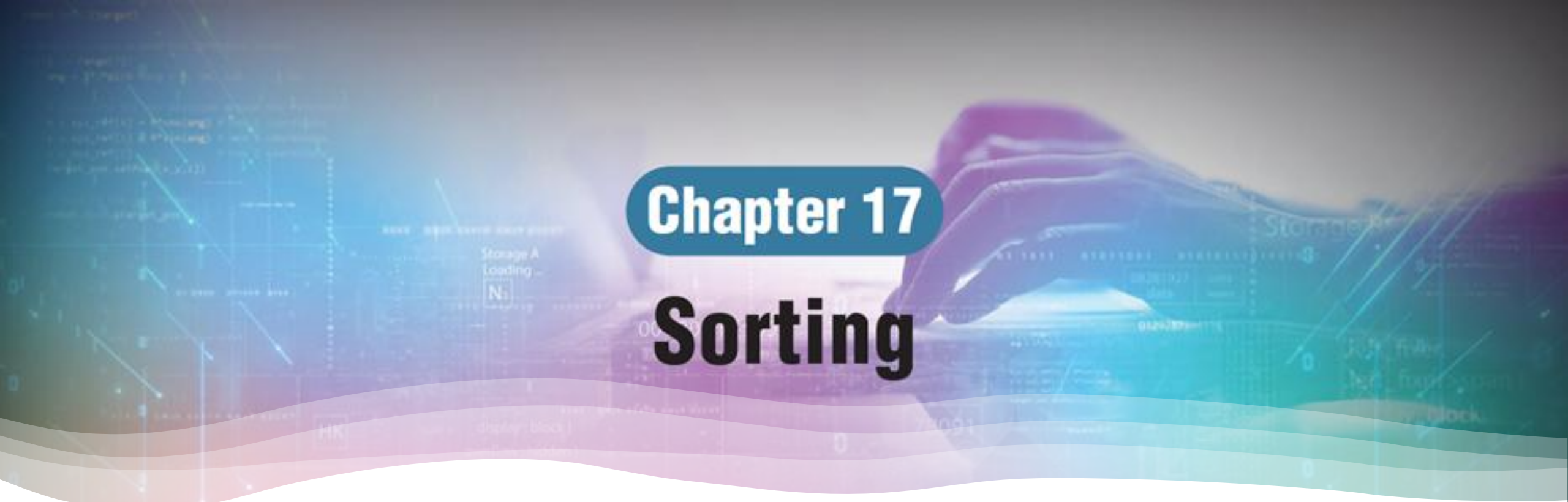




JAVA OOP AND DATA STRUCTURES

WITH INTRODUCTION TO SECURE CODING

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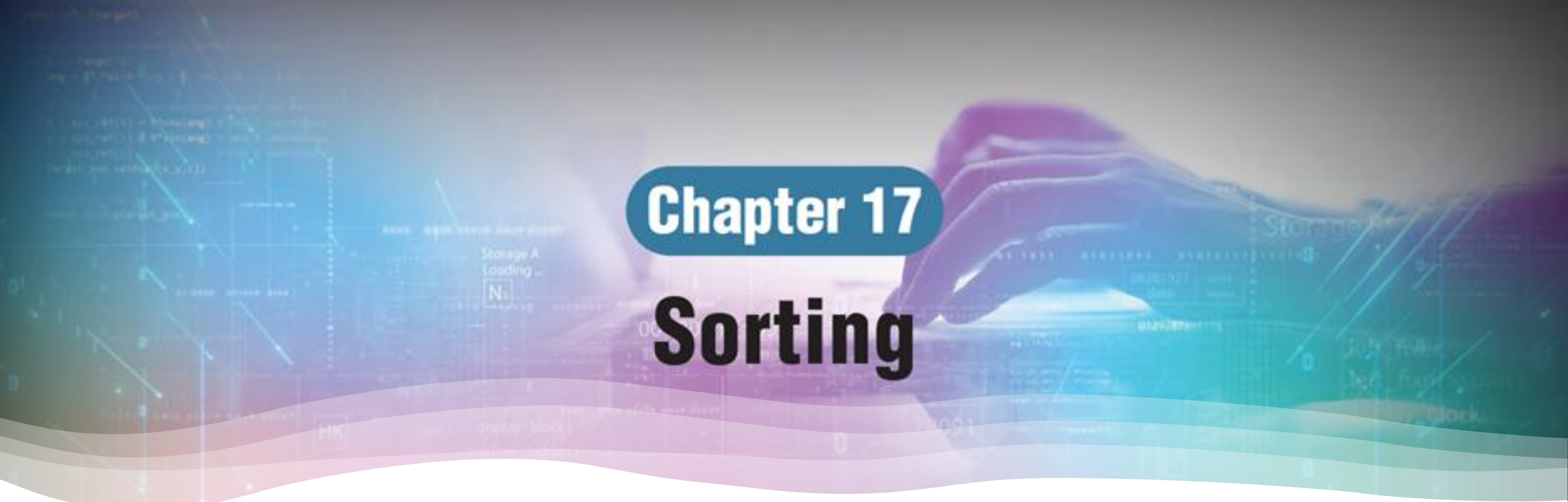
Chapter 17

Sorting

Chapter 17 Sorting

Java OOP and Data Structures with Introduction to Secure Coding

Dr. Ziping Liu



Chapter 17

Sorting

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Chapter Outlines

- Bubble Sort
- Selection Sort
- Insertion Sort
- ShellSort
- QuickSort
- MergeSort



17.1 Bubble Sort

- Bubble sort, also known as exchange sort, is an iterative algorithm that repeatedly traverses a list of unsorted elements (such as an array). It compares adjacent pairs of elements and swaps them if they are out of order. This process continues until no further swaps are needed, indicating that the list is sorted.
- Implementation of bubble sort requires a nested loop structure. The outer loop runs as long as swaps are necessary, while the inner loop iterates through the unsorted portion of the list, performing comparisons and swaps.
- With each iteration of the outer loop, the largest element in the working sublist "sinks" to its correct position at the end. After each pass, the largest element is sorted and excluded from further comparisons. This process repeats until only one element remains in the inner loop, at which point the list is fully sorted.



17.1 Bubble Sort

- Let's discuss how the list of numbers are sorted:
 - 69, 35, 18, 87, 50, 58, 79



17.2 Selection Sort

- Selection sort is an iterative algorithm that sorts a list by placing one element at a time in its correct position. The algorithm uses a nested loop structure: the outer loop tracks the current position to be filled, and the inner loop identifies the best element—typically the smallest or largest in the remaining unsorted portion—to place in that position.



17.2 Selection Sort

- Let's discuss how the list of numbers are sorted:
 - 69, 35, 18, 87, 50, 58, 79



17.3 Insertion Sort

- Insertion sort is an iterative algorithm that divides the list into a sorted sublist and an unsorted sublist. The algorithm repeatedly inserts elements from the unsorted sublist into their correct positions in the sorted sublist, one by one. It requires a nested loop: the outer loop keeps track of the element to be inserted, while the inner loop searches for the correct position to insert that element.



17.3 Insertion Sort

- Let's discuss how the list of numbers are sorted:
 - 69, 35, 18, 87, 50, 58, 79







17.4 Shell Sort

- Shell sort is a variation of insertion sort that incorporates an adjustable step (or gap) size when comparing the key element with its predecessors. The step size typically starts as half the size of the list and decreases by half in subsequent iterations until it reaches 1.
- The algorithm involves three tiers of nested loops:
 - The outermost loop iterates over the step sizes.
 - The middle loop progresses through the list, advancing the key element one by one.
 - The innermost loop compares the key element with its predecessors, located at a distance defined by the current step size, and adjusts their positions accordingly.



17.4 Shell Sort

- By starting with a larger step size, Shell sort allows out-of-order elements to be quickly moved closer to their final positions, making the subsequent iterations with smaller step sizes more efficient. This iterative algorithm effectively reduces the number of comparisons and swaps compared to traditional insertion sort, particularly for large datasets.



17.4 Shell Sort

- Let's discuss how the list of numbers are sorted:
 - 69, 35, 18, 87, 50, 58, 79



17.5 Quicksort

- Quicksort follows a divide-and-conquer strategy and can be done recursively. It begins by selecting a pivot element and then splits the list into two sublists: a left sublist where all elements are smaller than the pivot, and a right sublist where all elements are greater than the pivot. Each of these sublists is then further divided using the same process, with a pivot chosen from each sublist, until each sublist contains only one element.



17.5 Quicksort

- Let's discuss how the list of numbers are sorted:
 - 75, 95, 26, 39, 43, 62, 87, 23, 85, 57, 65, 30



17.6 Merge sort

- Merge sort follows a divide-and-conquer strategy and can be done recursively. Instead of dividing the list around a pivot point, merge sort splits the list into two sublists around the middle point. The process continues, with each sublist being split further until only one element remains in each sublist. After that, the sublists are merged in a bottom-up fashion into sorted sublists, which are further combined to form a sorted list of the original size.



17.6 Merge sort

- Let's discuss how the list of numbers are sorted:
 - 75, 95, 26, 39, 43, 62, 87, 23, 85, 57, 65, 30



Which national park is it?

- New River Gorge National Park

